Voxel-based compartmental models of toric spines: methods and validation

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Summary

The voxel method builds a compartmental model straight from the EM surface: each 0.5 µm voxel inside the cell is a compartment, neighbours are coupled by the cytoplasm between them, and the mesh's membrane is assigned to the voxels it encloses. Tested against analytic solutions and 3D finite elements, it is accurate for the soma and thick processes but cannot resolve toric-spine necks, which therefore stay as cables. Those cables, from mascaf, match the mesh in area and volume but not in resistance.

- **Simple shapes:** sphere, cylinder and ball-and-stick within 0.1–1.5% of closed-form solutions; spines attached to a voxel soma within 1% for input resistance, transfer and EPSP peaks.
- **The reconstructed neuron:** soma input resistance within 0.1% of FEM at every voxel size from 1 to 0.25 µm; thin dendrite tips +10% at 0.25 µm; the reduced one-compartment soma with access resistances within 0.1–0.6% of the full voxel model.
- **Toric spines:** mascaf cables overstate neck resistance by a median of 49% (range −31% to +140%) at 28 FEM-checked sites. Rescaling each segment to the mesh cross-section brings the median to −4%. A cable built from fine voxels underestimates resistance at necks thinner than about 0.15 µm (median −37%), as do voxel-only spines.
- **Consequence:** the simulations so far overstate how strongly toric spines are compartmentalized; with mesh cross-sections the median in vivo AZ→soma transfer is 0.66 rather than 0.54.

Background: two ways from a mesh to a model

There are two routes from an EM surface mesh to an electrical model: reduce the shape to a 1D skeleton with radii (a cable model), or discretize the 3D volume directly. mascaf takes the first route; the voxel method here takes the second, as do published 3D neuron solvers.

Cable models from meshes: MCF skeletons and mascaf

- mascaf (Mesh and Skeleton Cable Fitting; Fox, Fischer, DeBello and Peña, preprint 2026) fits a cable model to a closed surface mesh and handles cyclic topology, which motivated it: toric spines have high curvature, dense branching and topological holes.
- The skeleton comes from mean-curvature-flow (MCF) skeletonization (Tagliasacchi et al. 2012; CGAL): the surface is contracted by an area-minimizing flow toward a curve, pulled toward the medial axis by Voronoi poles. The input must be closed and a single connected component.
- Radii are set from mesh cross-sections; the default is equivalent area, $r = \mathrm{sqrt}(A/\pi)$, with perimeter, median, circle-fit and nearest-surface alternatives.
- mascaf's validation compares total volume, total surface area and containment against the mesh. It does not compare electrical quantities, which is the gap this document addresses.

Models on the 3D geometry

- The cable equation is the first member of a hierarchy of 1D equations derived from the 3D problem; the next member adds taper and transverse potential gradients (Lindsay et al. 2004).
- The EMI model (Tveito et al. 2017) solves intracellular, membrane and extracellular domains on a 0.5 µm Cartesian grid, the closest published precedent to a voxel cell. For a thick idealized cell, the cable equation differed from it by several mV.
- STEPS 4 computes membrane potential on tetrahedral meshes and matches the analytic Rallpack 1 cable to mean-square errors of 0.02–0.07 mV^2.
- The voxel method here is the intracellular-only special case of these approaches (extracellular space grounded), discretized by finite volumes on a cubic lattice. No published neuron model using exactly this construction was found (membrane from mesh triangles assigned to voxels, spines as cables); its parts are standard.

Junctions and lattice error

- A current entering a half-space through a disk of radius a meets a spreading resistance of $r_L/(4a)$ (Maxwell/Holm contact resistance); it is used here for neck-to-soma junctions.
- On an infinite simple cubic lattice of bond resistance R, the resistance from one node to infinity is $0.5055 \cdot R$ (Asad et al. 2009), the lattice counterpart of point-source spreading resistance. With $R = r_L/h$ this predicts the single-voxel self-term that the

hybrid junction adds (measured here as 0.34·r_L/h at a surface voxel).

- Electrodiffusion models predict non-Ohmic neck resistance at large currents (Cartailler & Holcman); at synaptic currents of 10–100 pA the effect is small, and all models here are Ohmic.

Methods: building a compartmental model from voxels

The voxel method turns a closed EM surface mesh directly into a compartmental model: every cubic voxel inside the cell is one compartment, neighbouring voxels are coupled by the axial conductance of the cytoplasm between them, and the membrane of the reconstructed surface is assigned to the voxels it encloses. No skeleton or radius estimate is used. It is a cell-centred finite-volume discretization of the 3D volume-conductor equation for the cell interior, with the extracellular space grounded.

Continuous problem

Inside the cell the potential obeys current conservation; at the membrane the outward current is capacitive plus leak (plus any channel or synaptic current):

```
\nabla \cdot (\sigma \nabla V) = 0 \quad \text{in } \Omega, \qquad -\sigma \,
\partial_n V = c_m \, \partial_t V + g_{pas} (V - E_L) + i_{ion} \quad
\text{on } \partial\Omega
```

with sigma = 1/r_L. The cable equation is the 1D limit of this problem for thin processes; the voxel method solves it without that limit.

Step 1. Mesh preparation

- Input: the closed EM surface (here `cell_wrapped_simplified.obj`, 330,326 triangles, scaled 0.005 µm per pixel). Self-intersections are repaired with pymeshfix where needed (84 faces changed, area and volume unchanged).
- Requirement found during validation: triangles must be small compared with the voxel size, because membrane is assigned triangle by triangle. The cell mesh has a median longest edge of 0.25 µm (99% < 0.67 µm), adequate for h = 0.5 µm.

Step 2. Rasterization

- The bounding box is divided into cubes of edge h (0.5 μm in the cell model; 0.25–1 μm in tests). A voxel belongs to the cell when its centre is inside the surface (trimesh voxelization with interior fill, or slice-by-slice point-in-polygon tests).
- Only the largest face-connected component is kept. Islands that touch the cell only at an edge or corner would otherwise carry membrane but no conducting path (this raised the soma input resistance from 78.8 to 102 MΩ in a first test).
- In the hybrid cell model, voxels inside the nine toric-spine meshes are removed; those spines are represented by cables (Step 5).

Step 3. Axial conductances

Two face-neighbour voxels exchange current through a square face of area h^2 over a centre-to-centre distance h, so the coupling conductance is

```
g_{ij} = \sigma \, \frac{h^2}{h} = \frac{h}{r_L}
```

(0.5 μS for h = 0.5 μm and r_L = 100 Ω cm). This is the standard two-point flux approximation; it is second-order accurate in the interior of a smooth region and first order at the staircase boundary.

Step 4. Membrane

Each surface triangle's area A is assigned to the voxel whose centre is nearest the triangle's centroid. Voxel i then carries leak g_pas·A_i, capacitance c_m·A_i and any channels or synapses placed there. The total membrane area equals the mesh area exactly, whatever the staircase shape of the voxel set; only its placement is discretized (to within about h).

Step 5. Toric spines as cables (hybrid model)

- Each toric spine is the SWC skeleton from the mascaf/MCF pipeline. Each SWC edge is a frustum split into pieces of at most 0.25 μm; a piece between radii r_a and r_b over length l has axial conductance pi·r_a·r_b/(r_L·l) and lateral area pi(r_a + r_b)·sqrt(l² + (r_b − r_a)²), split between its ends.
- Loops (the defining feature of toric spines) are closed exactly by merging the paired SWC samples that the skeleton duplicated to break each cycle.

- Each spine neck joins its nearest voxel through the spreading resistance of a disk of radius a on a half-space, r_L/(4a) (1.3 MΩ for the median neck radius of 0.19 µm, 3.8 MΩ for the thinnest, 0.066 µm).

Step 6. Equations and integration

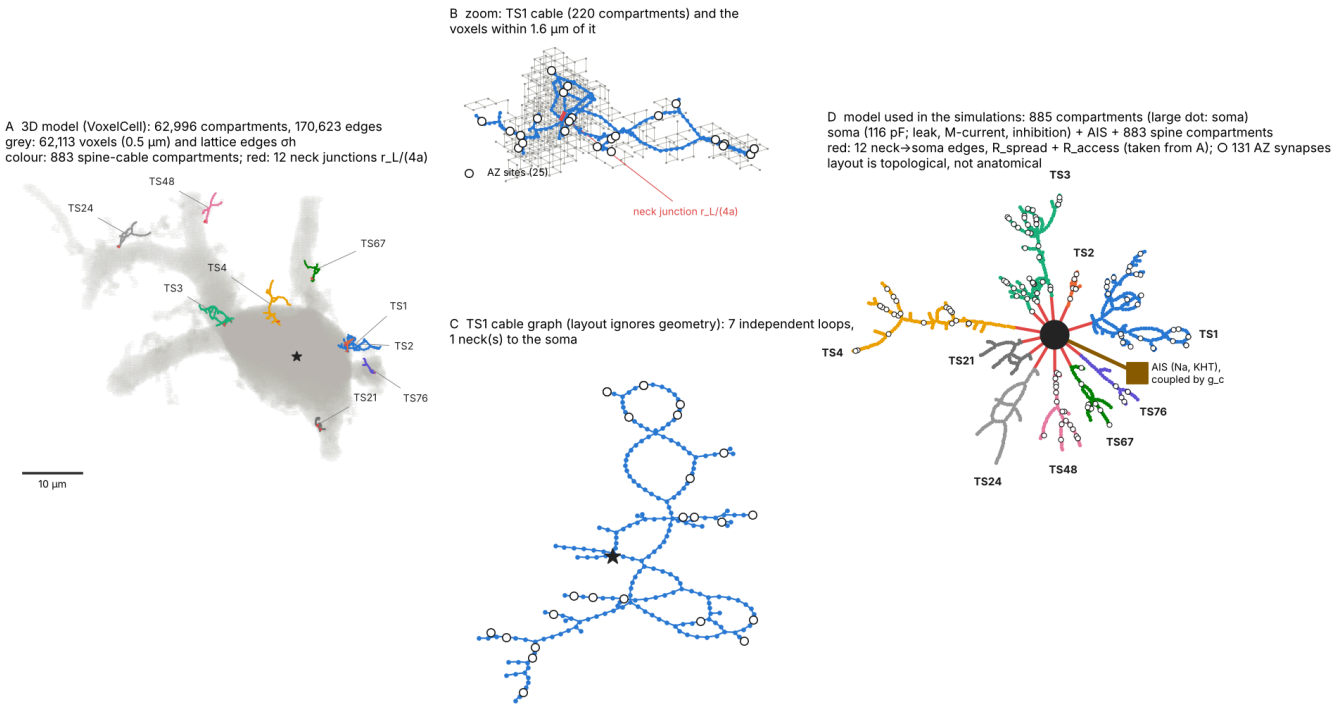
Collecting all compartments (voxels and cable nodes) gives one sparse linear system,

```
C \, \frac{dV}{dt} = -(L + G_m)(V - E_L) + I_{syn}(t) + I_{inj}(t)
```

where L is the graph Laplacian of the axial conductances, and C and G_m are diagonal. It is integrated by backward Euler (dt 0.01–0.05 ms) with a sparse LU factorization (SuperLU) reused at every step; steady states use the same factorization, or conjugate gradients preconditioned by algebraic multigrid for the largest systems. Synaptic conductances on a few nodes are handled by low-rank updates of the factorized system.

Step 7. Reduced models derived from the voxel model

- One-compartment soma with access resistances (used in the simulations): all voxel membrane is lumped into one compartment, and each spine neck joins it through r_L/(4a) + R_acc, where R_acc = R_in(attachment voxel) − R_in(whole cell) is measured in the voxel model. This is exact at steady state for single-site inputs.
- Soma plus dendrite cables: soma-labelled voxels form one compartment; dendrite voxels are binned by Euclidean geodesic distance from the soma (1 µm bins, each bin split into its connected pieces so that branches separate), giving cable compartments whose cross-section is the binned volume divided by the bin length.



Parameters throughout: r_L = 100 Ω cm, c_m = 1 µF/cm², g_pas = 2.63×10^{-4} S/cm² (in vivo high-conductance state) or 2.5×10^{-5} S/cm² (resting, slice).

Methods: reference solutions and tests

Every model was checked against a solution it does not share code or approximations with: closed-form results where they exist, and a 3D finite-element (FEM) solution of the same continuous problem otherwise.

Analytic references

- Sphere (isopotential): R_in = 1/(g_pas·4πr²); membrane time constant c_m/g_pas.
- Sealed cylinder of radius a and length L: V(x)/V(0) = cosh((L − x)/λ)/cosh(L/λ) and R_in = r_a·λ·coth(L/λ), with λ = sqrt(a/(2·r_L·g_pas)) and r_a = r_L/(πa²).
- Ball-and-stick: R_in = 1/(G_soma + G_cylinder), tip/soma ratio 1/cosh(L/λ).

3D finite elements

- Volume mesh: tetgen (quality bound 1.5–2.0) inside the same closed surface; 850,435 tetrahedra and 264,873 nodes for the cell, 50–70k tetrahedra for the test shapes.

- Discretization: linear (P1) elements (scikit-fem). The Robin membrane condition enters as a boundary mass term g_pas·u·v on the surface triangles; transients add c_m on the same boundary (lumped) with backward Euler.
- Sources and readouts: current injected uniformly into a small ball (radius 0.12–1.5 μm) around a site, and potentials averaged over balls; input and transfer resistances are these averages divided by the current.
- Solver: conjugate gradients with an algebraic-multigrid V-cycle preconditioner and a residual check (pyamg's own CG diverged on near-singular resting-membrane systems and was replaced).
- Convergence: coarsening the cell mesh to 531,696 tetrahedra changed spine-head input resistances by < 0.4% and transfer resistances by < 0.01%.

Test geometries

Test	Geometry	Reference	Models tested
T1	Sphere, r 8 μm	Analytic	Voxel (h 0.5, 1), FEM
T2	Sealed cylinder, r 1 μm, L 300 μm	Analytic cable equation	Voxel (h 0.25, 0.5), cable
T3	Sphere r 8 + cylinder r 1, L 300 μm	Analytic	Voxel, hybrid voxel soma + cable
T4	Sphere r 6 + neck (a 0.1, L 1 μm) + cylindrical head (r 0.3, L 0.6 μm)	FEM	Hybrid
T5	Sphere r 6 + neck (a 0.1, L 1 μm) + ring (centreline radius 1 μm, tube radius 0.2 μm)	FEM	Hybrid; voxel-only (h 0.05–0.2)

The test cylinders were meshed with rings every 0.25 μm so that their triangles are small (Step 1). Spine tests used a 50 pA alpha-function current (τ 0.5 ms) for transients.

Results: simple geometries

On shapes with known answers the voxel lattice reproduces somata and thick processes to within about 1%, and the hybrid model (voxel soma + cable spine) reproduces spine-head potentials and EPSP time courses to within 1%. Two exceptions are informative: thin

necks cannot be represented by voxels at any practical resolution, and the cable convention of running a thin neck to a ring's centreline overstates neck resistance by 12–15%.

Somata and thick processes

Test	Quantity	Model (h)	Error vs reference
T1 sphere	R_in	voxel (1.0 / 0.5 μm)	+0.08% / +0.14%
T1 sphere	membrane τ	voxel (0.5 μm)	+0.13% (backward-Euler step error)
T1 sphere	R_in	FEM	+0.04%
T2 cylinder	R_in	cable	0.00%
T2 cylinder	R_in	voxel (0.5 / 0.25 μm)	+0.29% / −0.73%
T2 cylinder	V(L)/V(0)	voxel (0.5 / 0.25 μm)	−0.99% / +0.65%
T3 ball-and-stick	R_in soma	voxel (0.5 / 0.25 μm)	+0.75% / +0.37%
T3 ball-and-stick	V(tip)/V(soma)	voxel (0.5 / 0.25 μm)	−1.5% / −0.14%
T3 ball-and-stick	R_in soma	hybrid (0.5 μm)	+0.47%

The sphere stays isopotential in the voxel model to within 0.1% across its volume. The voxel cylinder's residual error comes from its staircase cross-section, which is 4.5% too small at h = 0.5 μm and 3.5% too large at h = 0.25 μm for a 1 μm radius: the error does not shrink monotonically, because whether voxel centres fall inside a circle depends on alignment.

Spines on a soma

Test	Model	Spine-head R_in	Local R (head − soma)	Spine→soma transfer	EPSP peak, head
T4 stubby spine	hybrid, h 0.5	−0.01%	−1.6%	+0.07%	−0.03%
T5 toric spine	hybrid, neck to ring centreline, h 0.5	+0.59%	+13.1%	−0.63%	+0.83%
T5 toric spine	hybrid, neck to tube wall, h 0.5	+0.17%	+4.8%	−0.23%	+0.25%

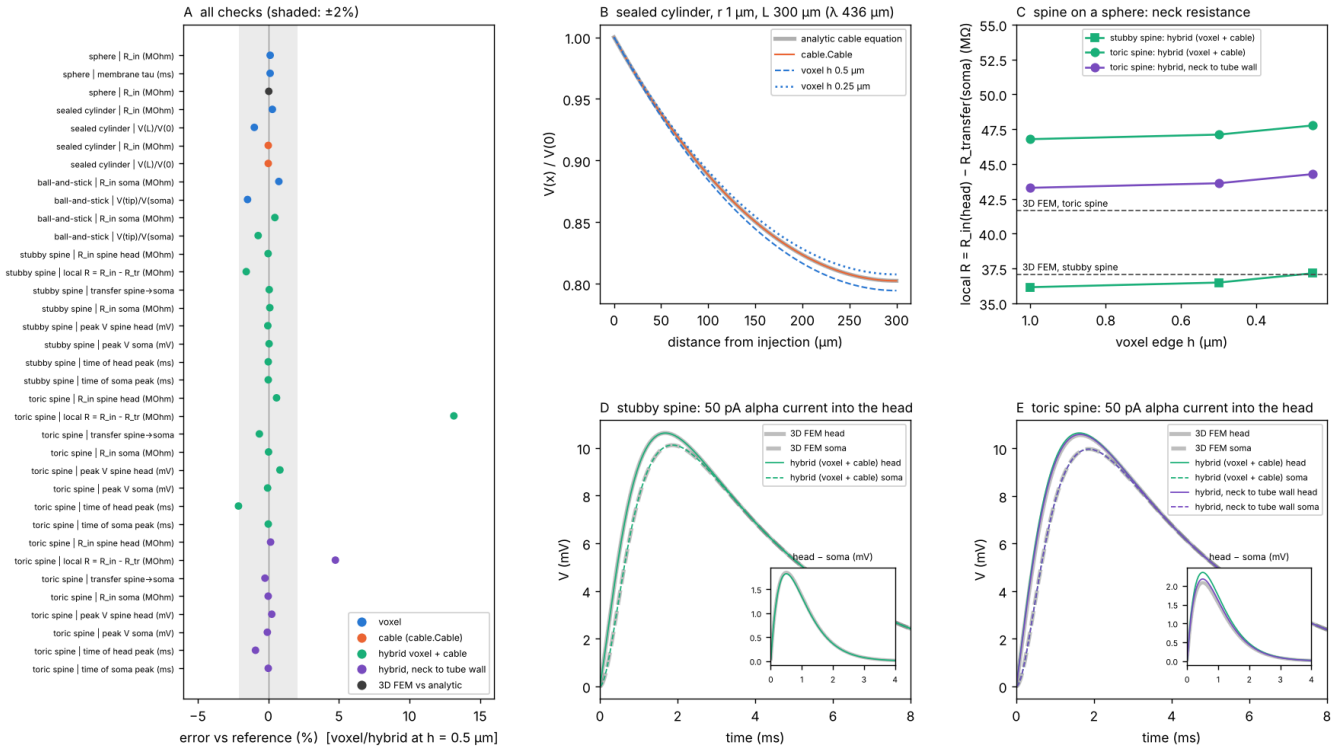
- Times of the EPSP peaks at the head and soma match FEM to within 0.035 ms; transfer and input resistances to within 0.7%.
- The toric spine's excess local resistance has a geometric cause. The skeleton carries the thin neck (radius 0.1 µm) all the way to the ring's centreline, 0.2 µm past where the neck actually meets the ring tube; that adds about 6 MΩ of thin-neck resistance. Stopping the neck at the tube wall leaves 4–6%, most of which is the lattice term below.
- The neck junction double-counts a little: the attachment voxel adds its own lattice self-resistance, $0.34 \cdot r_L/h$ (0.68 MΩ at h = 0.5 µm), on top of $r_L/(4a)$. This term grows as the voxels shrink, so the hybrid local resistance rises slightly with refinement (T5: 46.8, 47.1, 47.8 MΩ at h = 1, 0.5, 0.25 µm).

Voxel-only spines

With the spine itself voxelized (small soma, r 2 µm, so fine lattices fit), the local spine resistance converges poorly:

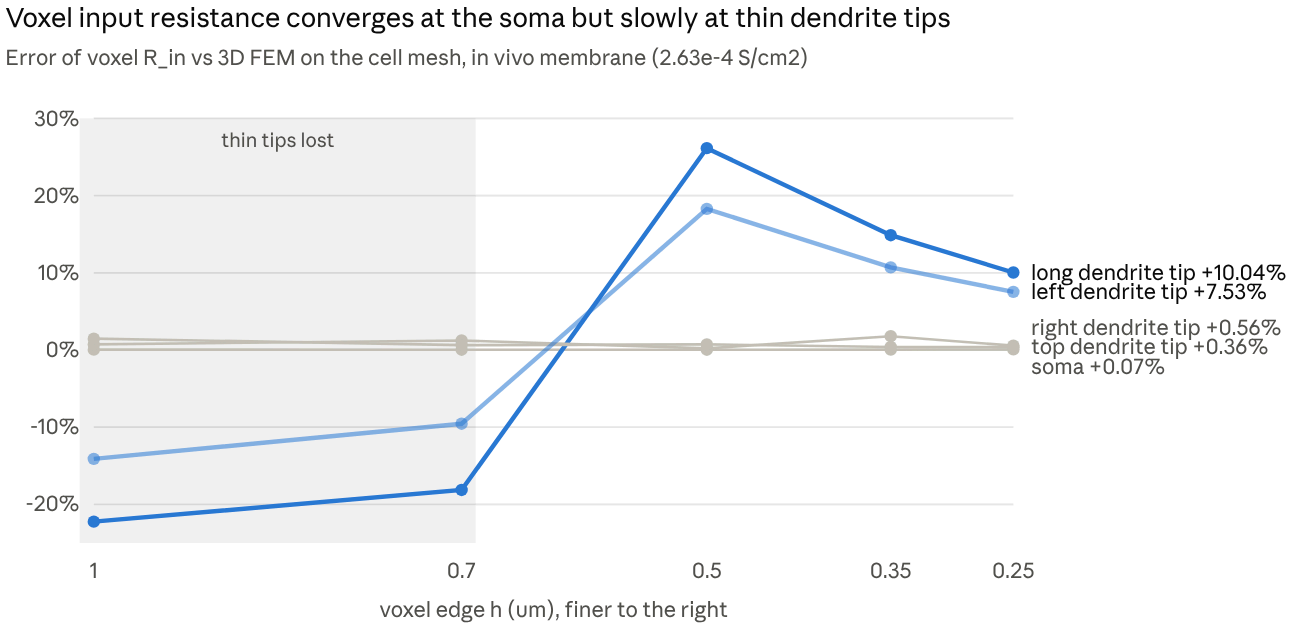
Spine	h 0.2 µm	h 0.1 µm	h 0.07 µm	h 0.05 µm
Stubby (neck a 0.1 µm)	spine lost	−14%	−16%	+7.3%
Toric (neck a 0.1 µm)	spine lost	−3.5%	−3.9%	+7.4%

At h = 0.2 µm the 0.2 µm-wide neck falls between voxel centres and the spine is cut off. At finer h the error is set by how many voxel centres fall inside the neck's circular cross-section (1–3 across), so it oscillates rather than shrinking. Representing toric-spine necks with voxels would need h well below 0.05 µm (millions of voxels per spine); cables are the practical choice for necks.



Results: the reconstructed neuron

On the full EM cell, the voxel model matches the 3D FEM solution at the soma to within 0.1% at every voxel size, so the somatic part of the model is accurate; thin dendrite tips converge slowly, and the toric spines cannot be voxelized at practical resolution.



results/validate_voxel_fem.csv · FEM 850,435 tetrahedra; voxels 5,844 (h 1) to 380,161 (h 0.25)

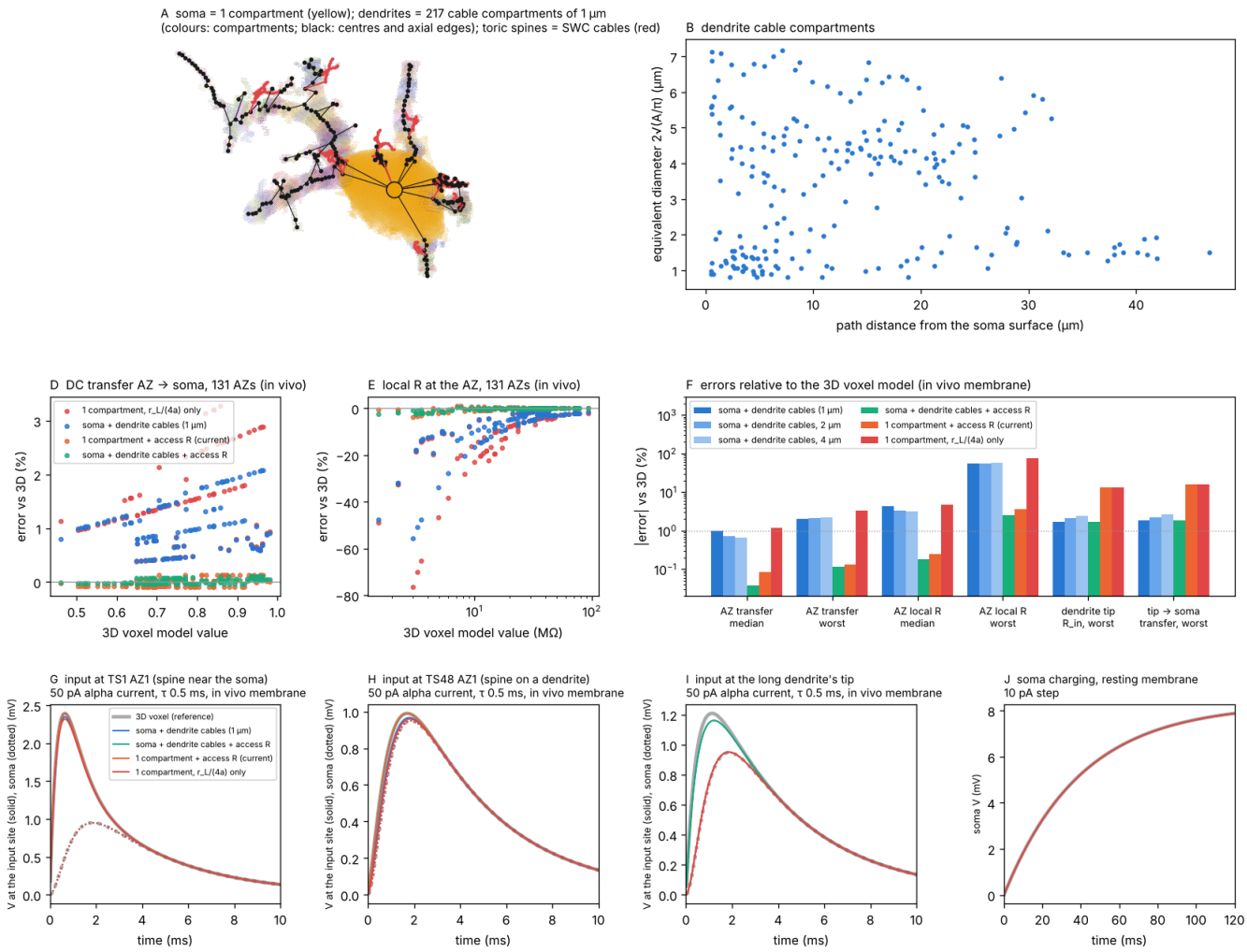
- Soma: 78.9 vs 78.8 MΩ (in vivo membrane) and 828.1 vs 828.1 MΩ (resting), with h from 1 to 0.25 μm. Transfer resistances from every site to the soma agree to within 0.1%.
- Thick dendrite tips (right, top): within 2% at all h.
- Thin tips (long, left, about 1 μm across): +26% and +18% at h = 0.5 μm, +10% and +7.5% at 0.25 μm. At h ⩾ 0.7 μm the tips are cut off by the rasterization and the site falls on the remaining stump. The resting membrane shows the same pattern with errors about 10 times smaller, because the cell is then far more compact.
- Toric spines in the voxel lattice: TS1 head 80.3 MΩ at h = 0.5 μm against 99.5 MΩ in FEM (the neck is not resolved); at h = 0.25 μm TS1 is close (98.4) but TS4 is 42% too high (165 vs 117 MΩ). This is the cell-scale version of the voxel-only spine test above.

Reduced models of the cell

Because the soma is effectively isopotential, the voxel cell can be collapsed with little loss. Errors relative to the 3D voxel model (in vivo membrane):

Model	Compartments	AZ→soma transfer, median / worst	Neck local R, median / worst	Dendrite tip R_in, worst
3D voxel soma + cable spines	62,996	reference	reference	reference
Soma + dendrite cables + access R	1,101	0.04% / 0.12%	0.18% / 2.6%	1.7%
Soma + dendrite cables	1,101	0.98% / 2.1%	4.4% / 56%	1.7%
One compartment + access R (used in the simulations)	884	0.08% / 0.14%	0.25% / 3.6%	13.5%
One compartment, r_L/(4a) only	884	1.2% / 3.3%	4.9% / 77%	13.5%

The access resistance (from the 3D model) carries the 3D spreading of current from a neck into the soma or a dendrite; without it, thick necks entering a dendrite (TS48, TS3) lose up to 56–77% of their local resistance. Somatic EPSPs agree within 0.6% in every model; dendritic cables matter only for voltages inside the dendrites (a tip EPSP is 21% too small in the one-compartment model).



Results: toric spines — voxel-derived cables versus mascaf cables

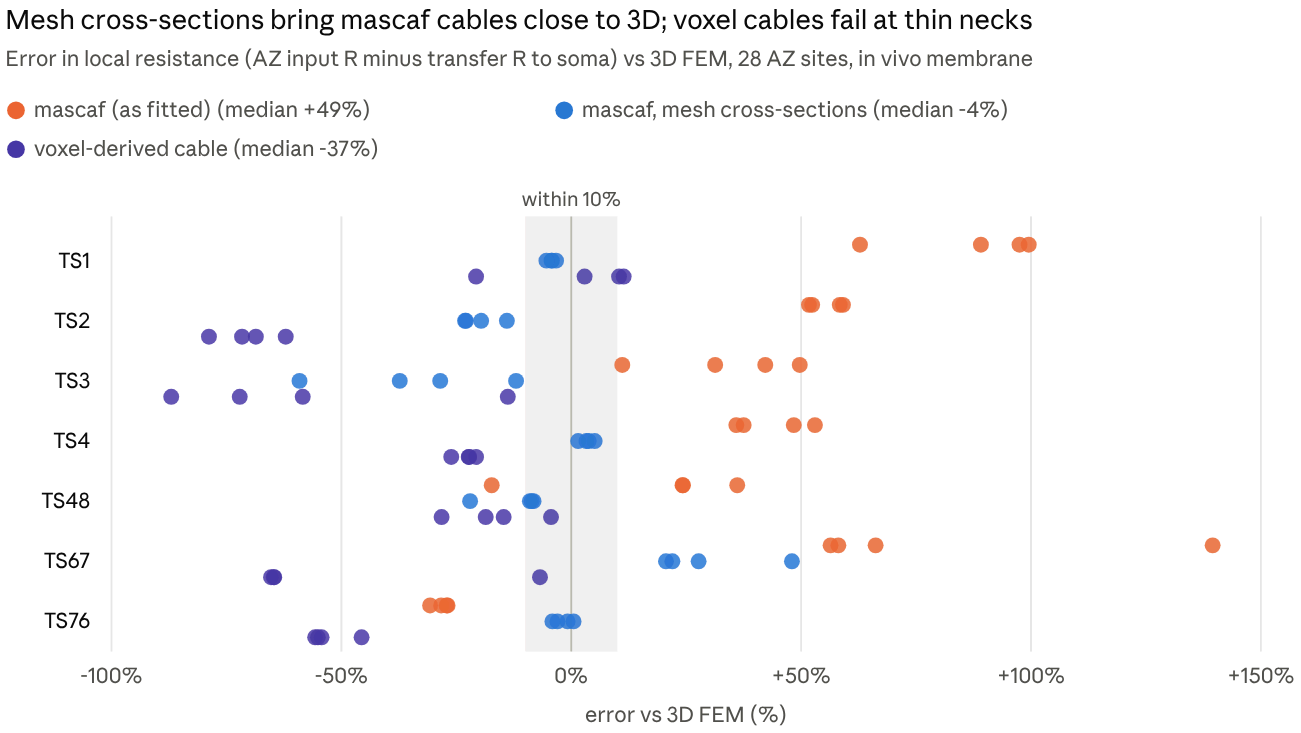
For the toric spines, mascaf's cable models get topology and total membrane right but overestimate neck resistance by a median of 49%. Rescaling each cable segment to the mesh cross-section at its midpoint brings the median error to −4%. A cable built directly from a fine voxelization fails at the thinnest necks, for the same reason the voxel-only spines failed.

Mesh used. These results use the repaired `cell_wrapped_simplified.obj` and the separate spine meshes (TS*.obj), the same inputs as the simulations. The new `cell_wrapped_simplified_1cc.obj` (soma and spines as one connected surface) was not reachable from this session because the computer holding the project folder was offline; the analysis scripts take the mesh as an argument and are ready to rerun on it.

Three ways to build the spine cables

- **mascaf as fitted**: MCF skeleton (pymcfs) plus mascaf radii with the equivalent-area strategy, r = sqrt(A/π) (the repository's default, `toric_spines_sim/geometry/mesh_pipeline.py`).
- **mascaf with mesh cross-sections**: same skeleton and nodes; each segment's axial conductance is rescaled so that its area equals the mesh cross-section cut perpendicular to the segment at its midpoint (`validate_neck_xsec.py`).
- **Voxel-derived cable** (`voxel_cable.py`): the cell surface is rasterized at h_s = 0.05–0.08 µm in a box around each spine; voxels inside the spine mesh are binned by geodesic distance from the soma (bins 0.125–0.5 µm); each bin's connected pieces become compartments (so rings stay rings); cross-section = piece volume / bin length; membrane = mesh triangles assigned to the nearest spine voxel; the neck joins the soma lattice through r_L/(4a) with a from the first piece's cross-section.

Accuracy against 3D FEM



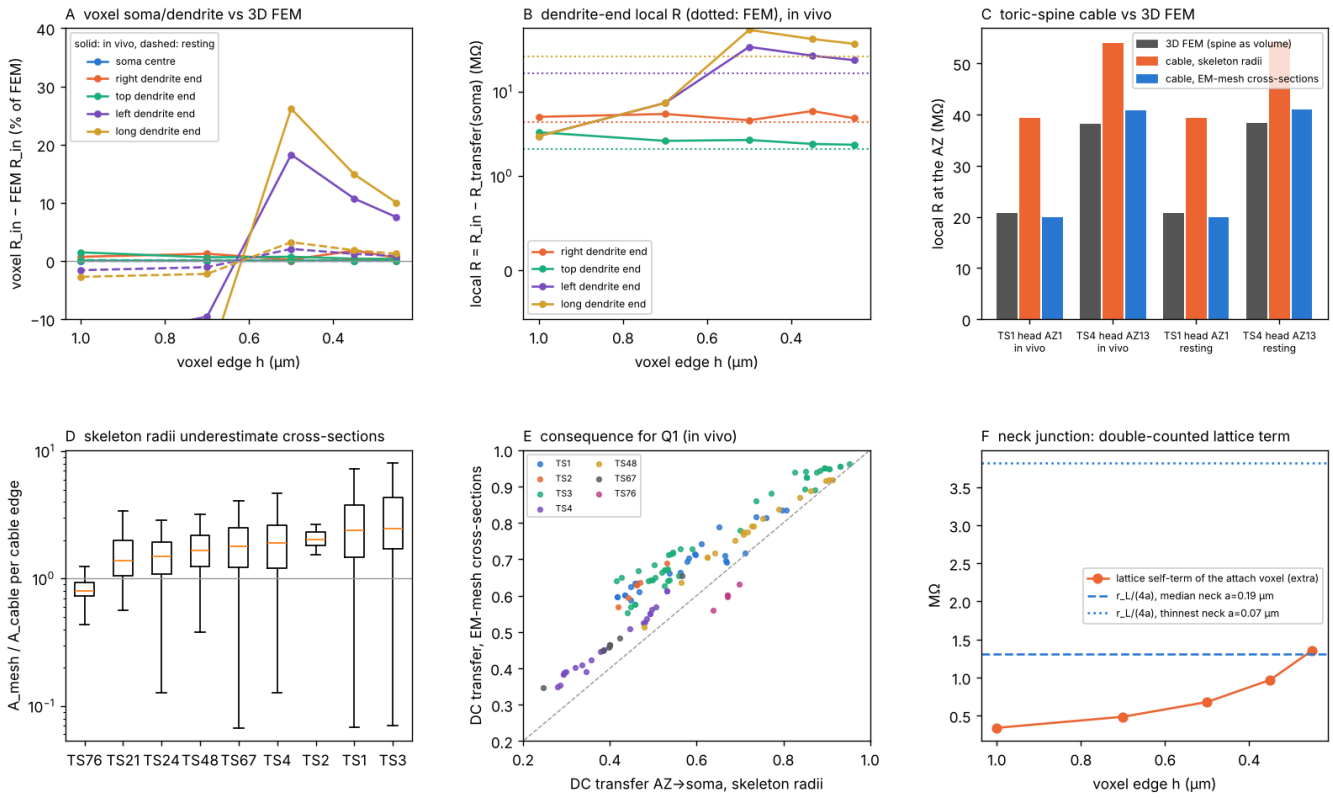
results/compare_spine_methods.csv · 28 AZ sites (up to 4 per spine) on the repaired cell mesh; FEM 850,435 tetrahedra

Model	Local R error, median	Range	Spines within 10%
mascaf as fitted	+49%	−31% to +140%	none
mascaf with mesh cross-sections	−4%	−59% to +48%	TS1, TS4, TS76 (TS48 within 10% at 3 of 4 sites)
Voxel-derived cable (h_s 0.05 µm, bins 0.25 µm)	−37%	−87% to +11%	TS1 at 3 of 4 sites
Voxel-derived cable, other settings (h_s 0.08; bins 0.125, 0.5 µm)	−35% to −40%	−89% to +21%	same pattern
All-voxel cell, h 0.25 µm	+13%	−92% to +460%	none

The all-voxel lattice either cuts spines off or bridges them to the soma, depending on where the 0.25 µm voxel centres fall.

Why mascaf's cables are too resistive

- **Geometry checks pass.** Summed over each spine, the cable′s membrane area is 93–99% of the mesh area and its volume 97–125% of the mesh volume (watertight spines). These are the quantities mascaf′s own validation compares.
- **Resistance does not.** Axial resistance depends on ∫ dl/A, dominated by the narrowest stretches. Measured perpendicular to the skeleton at each segment midpoint, the mesh cross-section is a median 1.4–2.5× the cable′s (TS76: 0.8×). Area and volume can match while thin stretches are too thin and thick ones too thick.
- **Likely causes, not yet isolated.** Radii are set at nodes spaced 0.5–1 µm apart and interpolated linearly; nodes tend to sit at constrictions and junctions; and optional rescaling to match total surface area would shrink radii where the skeleton path is longer than the true axis. Which of these dominates has not been tested.



Why the voxel-derived cable fails at thin necks

The thinnest cross-section in every spine is 0.06–0.13 µm in radius (mesh-equivalent), only 1.5–5 voxels across at h_s = 0.05–0.08 µm. Counting voxels cannot measure such a section to better than about ±50%, and the first-bin neck radius comes out at 0.29–1.1 µm, far wider than the true neck. Spines whose resistance is dominated by such a neck (TS2, TS3 sites near the soma, TS67, TS76) come out 46–87% too low; spines with longer thick paths (TS1, TS4, TS48) are within −28% to +11%. Refining h_s from 0.08 to 0.05 µm did not change this.

Discussion: which method for which part

Use voxels for the soma and thick dendrites, cables for the spines, and check every cable against 3D before trusting its neck resistance.

Part of the cell	Recommended representation	Evidence
Soma, thick dendrites (≥ 1 µm across)	Voxels, h = 0.5 µm (or one compartment + access R)	Within 0.1% of FEM at the soma; within 2% at thick tips

Part of the cell	Recommended representation	Evidence
Thin dendrite tips	Voxels at h ⩽ 0.25 µm, or dendrite cables from binned voxels	+10% at h = 0.25 µm; cables within 2% of the voxel model
Spine necks and rings	Cables, with cross-sections measured on the mesh	mascaf + mesh cross-sections: median −4%, worst −59% / +48%
Neck-to-soma junction	r_L/(4a), minus the lattice self-term 0.34·r_L/h	+1–5% bias if the self-term is left in

What this means for the current results

- The simulations so far used mascaf radii, so they overstate neck resistance (median +49% at the 28 FEM sites, up to +140%) and therefore overstate compartmentalization. With mesh cross-sections, the in vivo AZ→soma transfer rises from a median of 0.54 to 0.66 (d_xsec_q1.py).
- The soma and the reduction to one compartment with access resistances are not in question: both agree with 3D to within 0.1–0.6%.

Limits of the voxel method

- **Thin processes.** A section must be several voxels across to be measured; necks of radius 0.06–0.13 µm would need h < 0.03 µm. Voxel counts also misjudge cross-sections non-monotonically, so refining h does not guarantee improvement.
- **Fusion and cut-off.** Rasterization connects surfaces closer than h (spines lying on the soma) and disconnects necks thinner than h. Both happened here; FEM, which meshes the surface itself, has neither problem.
- **Membrane placement.** Mesh triangles must be smaller than h; large triangles concentrate membrane in the wrong voxels.
- **Boundary accuracy.** The staircase boundary is first-order accurate. A cut-cell finite-volume scheme (face areas and volumes clipped by the surface) would remove the cross-section error; this was not tried.

Limits of the cable route

- Agreement in area and volume (mascaf's checks) does not imply agreement in axial resistance: here both matched within 7% and 25% while resistance was 1.4–2× too high.
- Where a thin neck meets a ring or the soma, a cable runs the thin radius to the skeleton junction; in the T5 test this alone added 12–15% to the neck resistance.
- Even with mesh cross-sections, TS3 (three necks, sites within 5 MΩ of the soma) and TS67 stay 20–60% off. These are the spines whose geometry is least like a tube.

Next steps

- Rerun `fem_az_sites.py` and `compare_spine_methods.py` on `cell_wrapped_simplified_1cc.obj` once it can be read from the project folder; a single connected soma-plus-spine surface removes the segmentation boundary between TS*.obj and the cell mesh, which affects TS76 and TS2.
- Find which mascaf step thins the cables (node placement, interpolation, or area rescaling) by refitting TS1 and TS4 with radii at denser nodes.
- Rerun Q1–Q4 with corrected spine cross-sections.

References and code

- Fox, Fischer, DeBello, Peña (2026). Robust cable models from complex neuronal morphologies (mascaf). Research Square preprint. Documentation; validation module.
- Tagliasacchi, Alhashim, Olson, Zhang (2012). Mean curvature skeletons. Comput Graph Forum 31:1735–1744. CGAL implementation.
- Lindsay, Rosenberg, Tucker (2004). From Maxwell's equations to the cable equation and beyond. Prog Biophys Mol Biol 85:71–116.
- Tveito et al. (2017). Accuracy of classical models for the membrane potential of neurons, compared with the EMI model. Front Comput Neurosci 11:27.
- Chen et al. (2022). STEPS 4.0. Front Neuroinform 16:883742.
- Asad et al. (2009). Two-point resistances of the infinite simple cubic resistor lattice. arXiv:0905.0056.
- Cartailler, Holcman. Electrodiffusion in dendritic spines. arXiv:1803.03940.
- Barn owl ICx toric spines, eNeuro 7(1) ENEURO.0197-19.2019.

Code (`analysis/compartmentalization/` in toric_spines_sim_claude): `cellmodel.py` (voxel cell, hybrid and reduced models), `validate_simple_shapes.py` and `validate_voxel_spines.py` (test shapes), `validate_voxel_fem.py` and `fem_az_sites.py` (3D FEM), `validate_neck_xsec.py` (cable vs mesh cross-sections), `voxel_cable.py` and `compare_spine_methods.py` (spine cable methods), `soma_dend_model.py` (soma + dendrite cables).